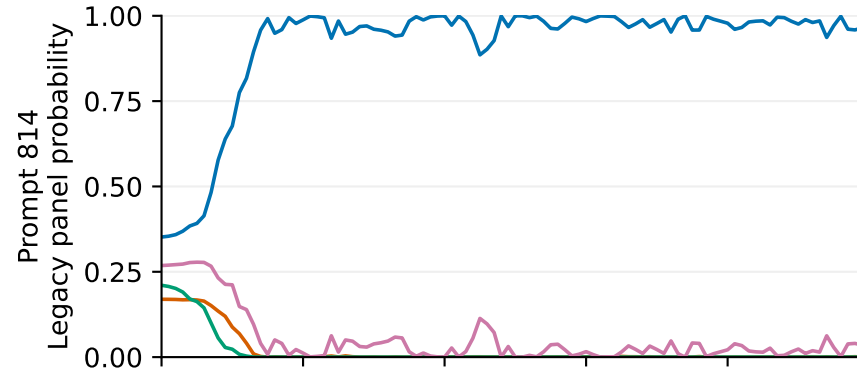


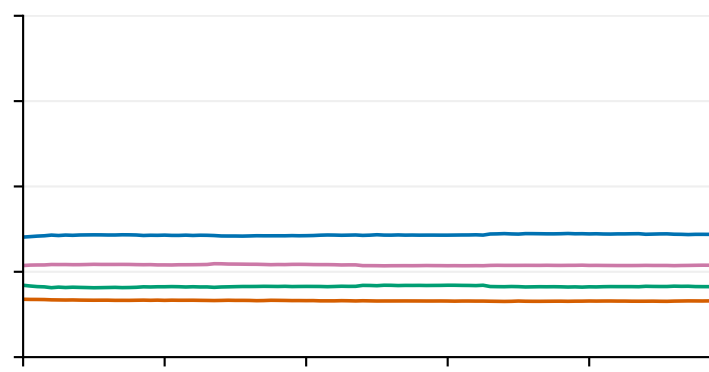
IPO | Original cyclic experiments

Legacy definition: $\pi_i = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π_i

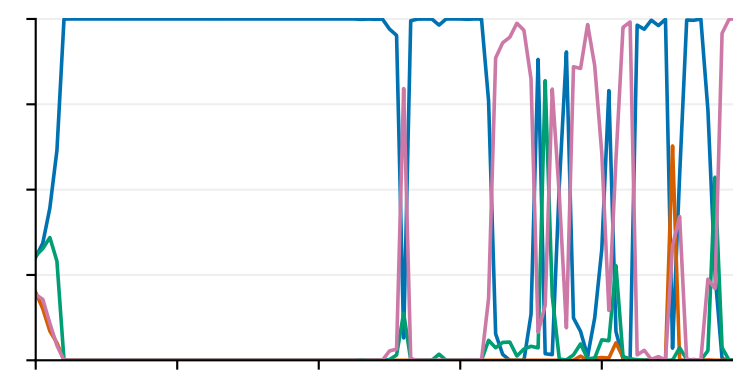
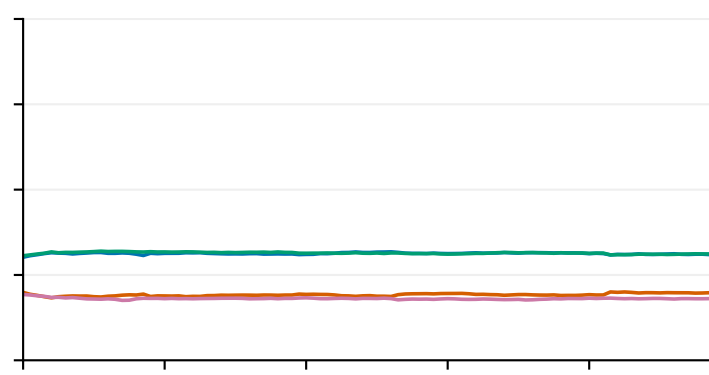
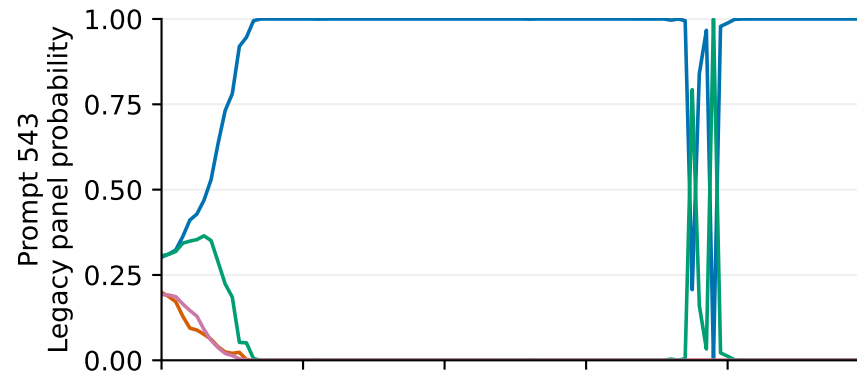
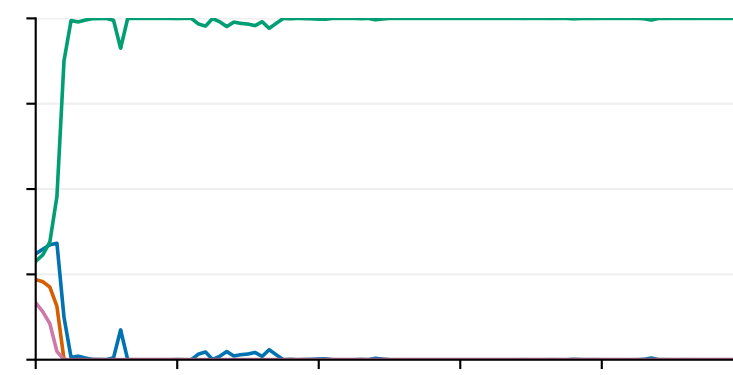
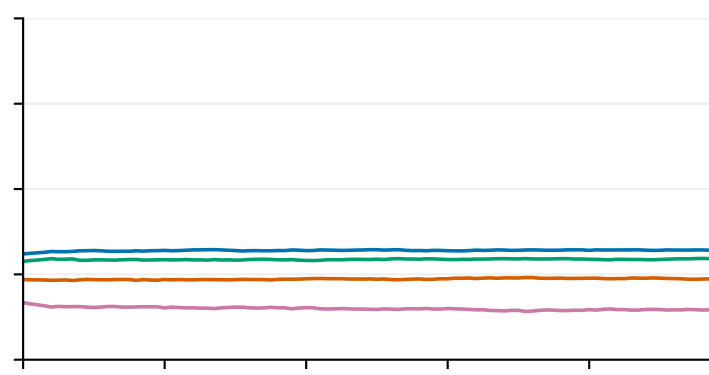
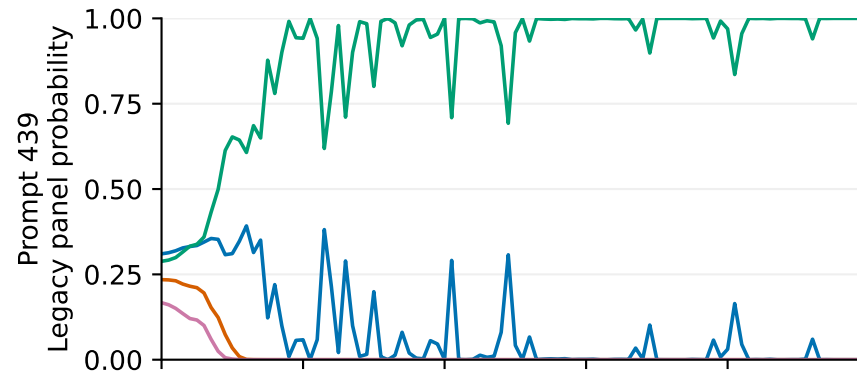
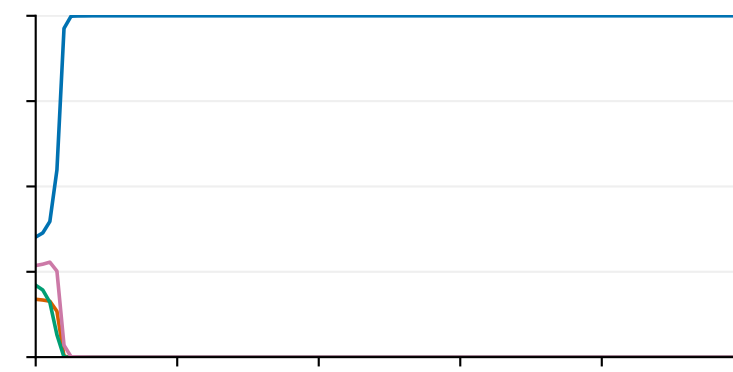
$\alpha=0.1, \lambda=0.1, \beta=0.01$



$\alpha=0.5, \lambda=0.5, \beta=5$



$\alpha=0.99, \lambda=0.8, \beta=1$



Outer iteration (pre-update)

Outer iteration (pre-update)

Outer iteration (pre-update)

Response 1

Response 2

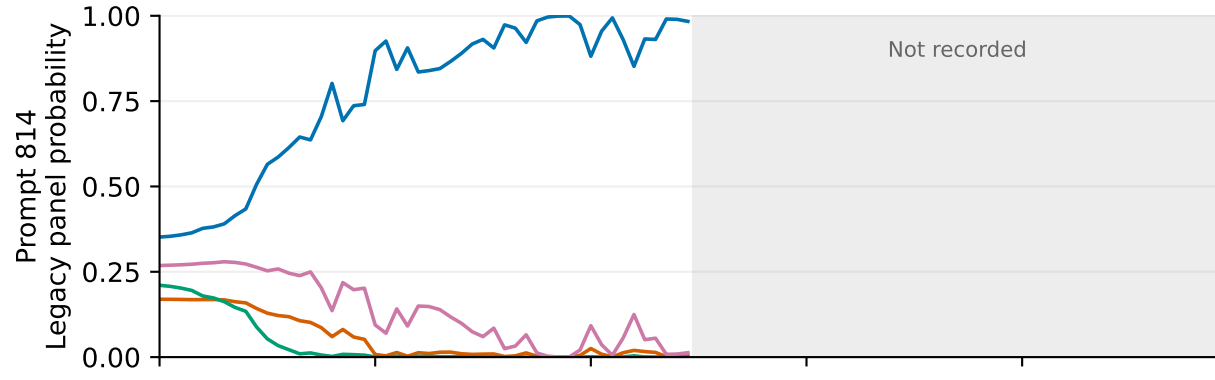
Response 3

Response 4

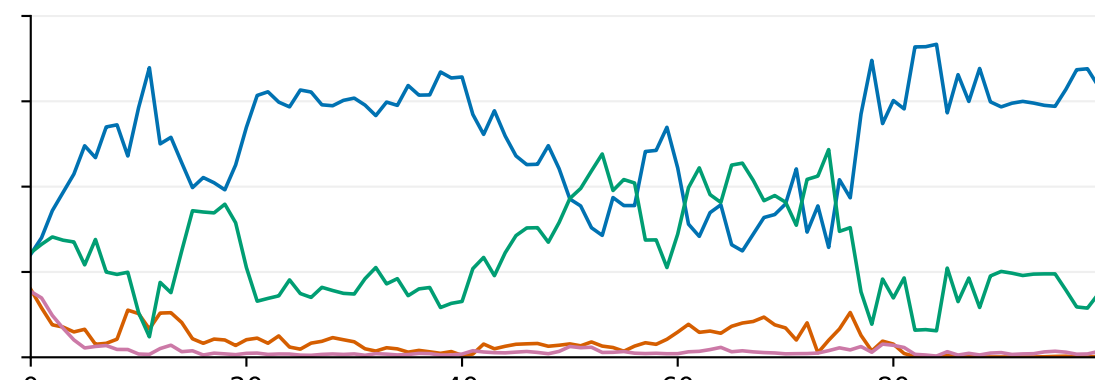
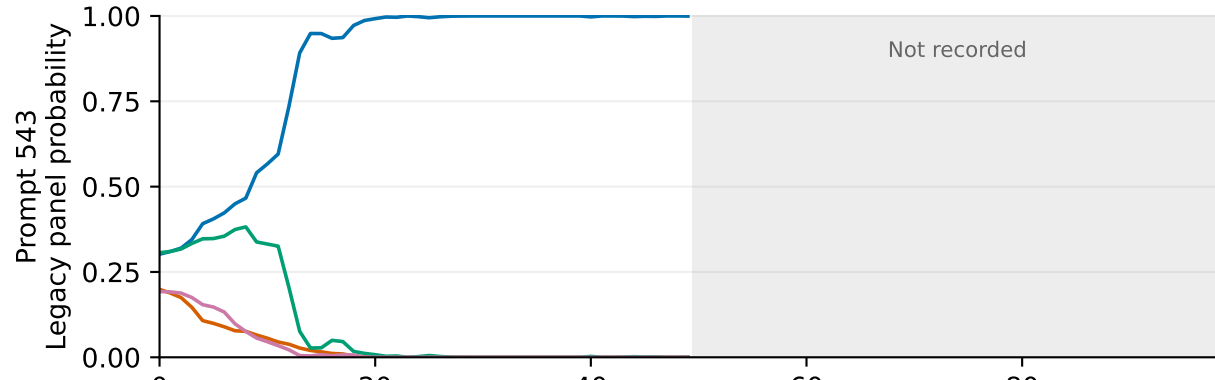
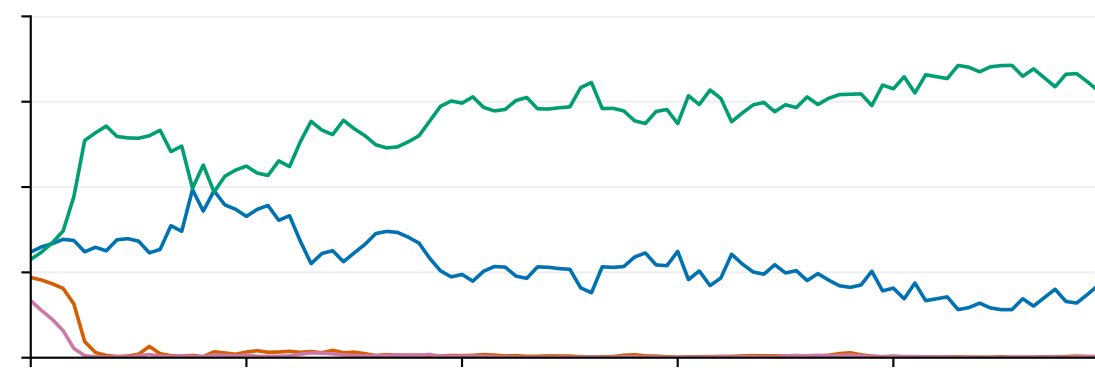
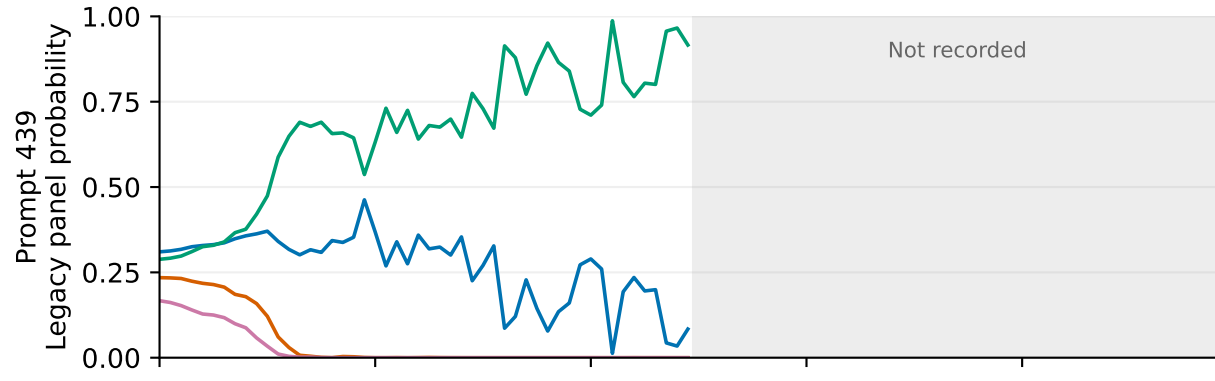
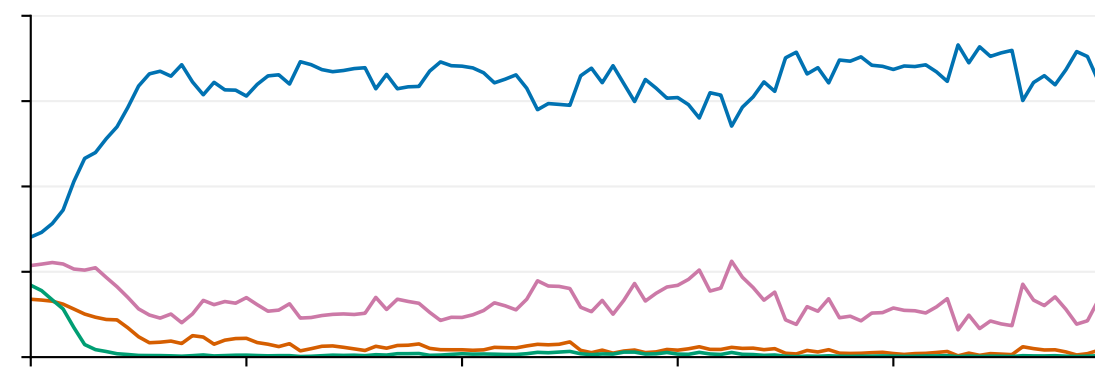
IPO | Original cyclic experiments

Legacy definition: $\pi = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π

$\alpha=0$, $\lambda=0$, $\beta=0.01$



$\alpha=0.99$, $\lambda=0.8$, $\beta=10$



Outer iteration (pre-update)

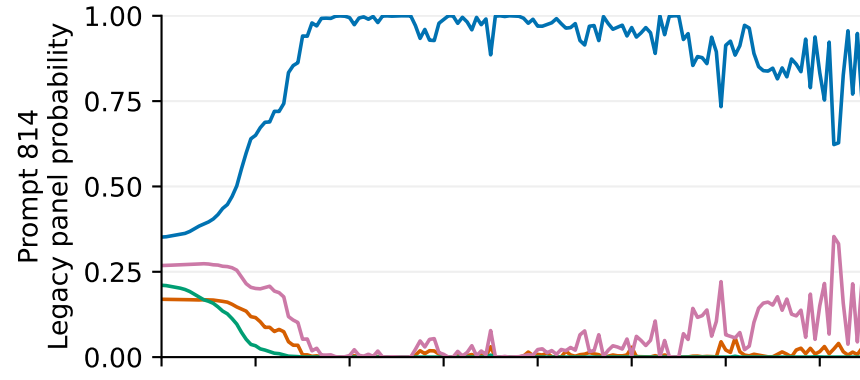
Response 1 Response 2 Response 3 Response 4

Outer iteration (pre-update)

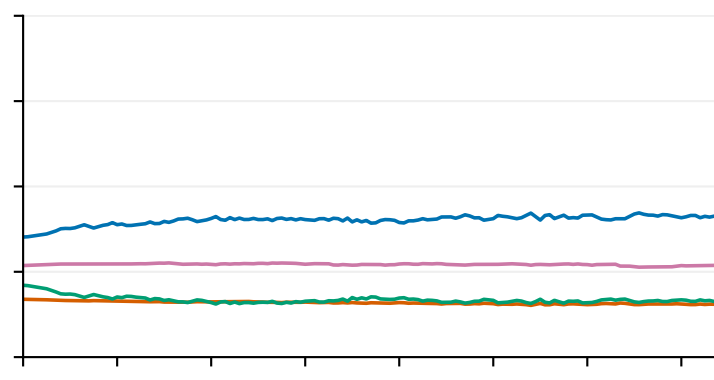
DPO | Original cyclic experiments

Legacy definition: $\pi_i = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π_i

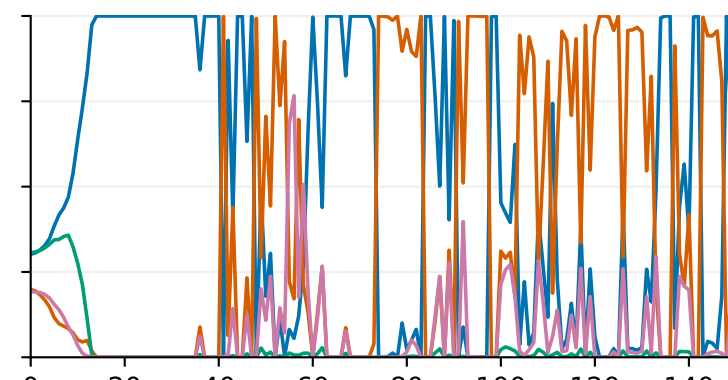
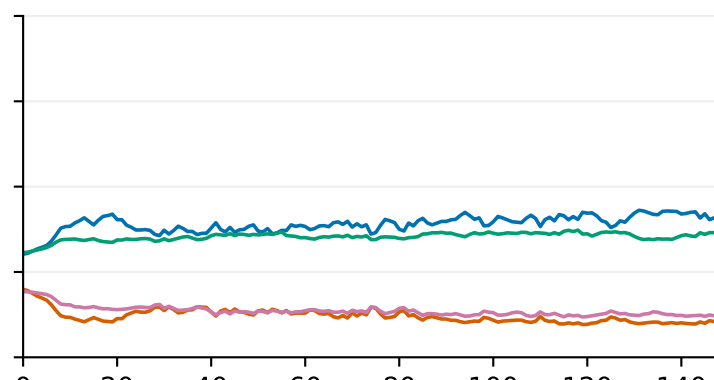
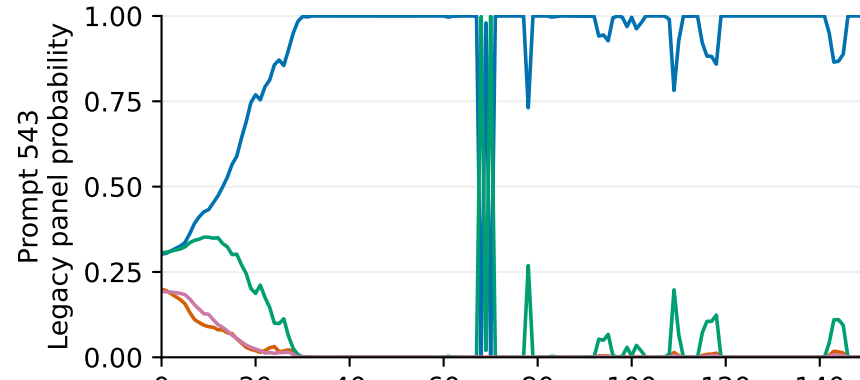
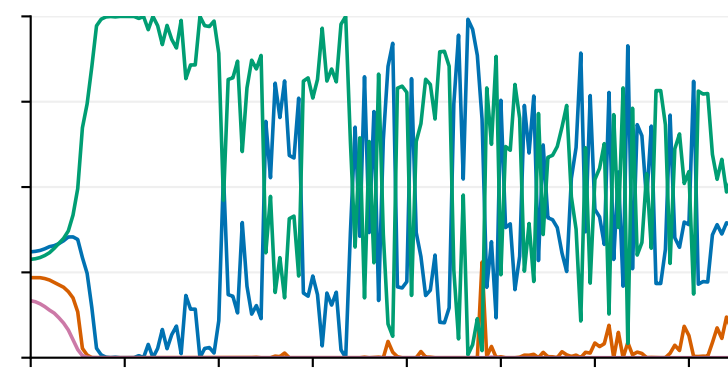
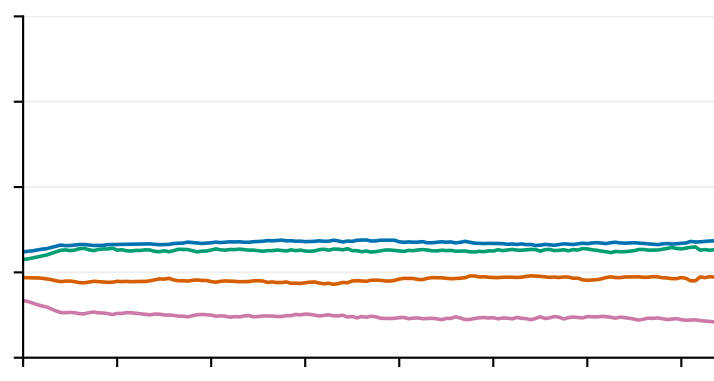
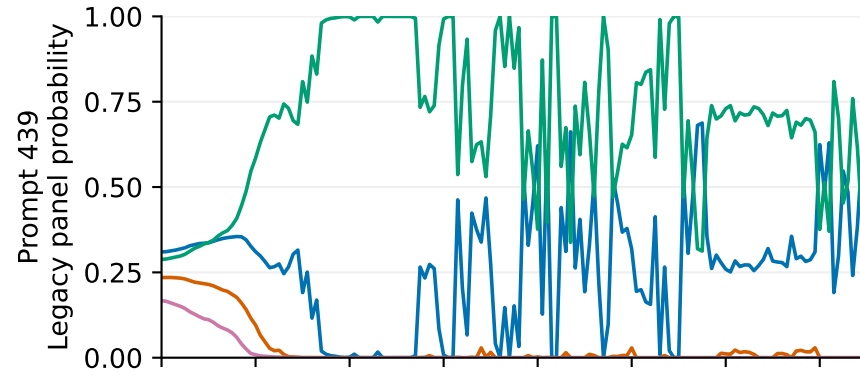
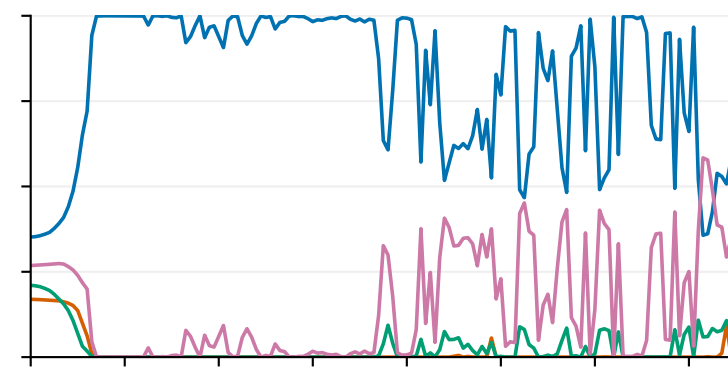
$\alpha=0.1, \lambda=0.1, \beta=0.01$



$\alpha=0.5, \lambda=0.5, \beta=5$



$\alpha=0.99, \lambda=0.8, \beta=1$



Outer iteration (pre-update)

Outer iteration (pre-update)

Outer iteration (pre-update)

Response 1

Response 2

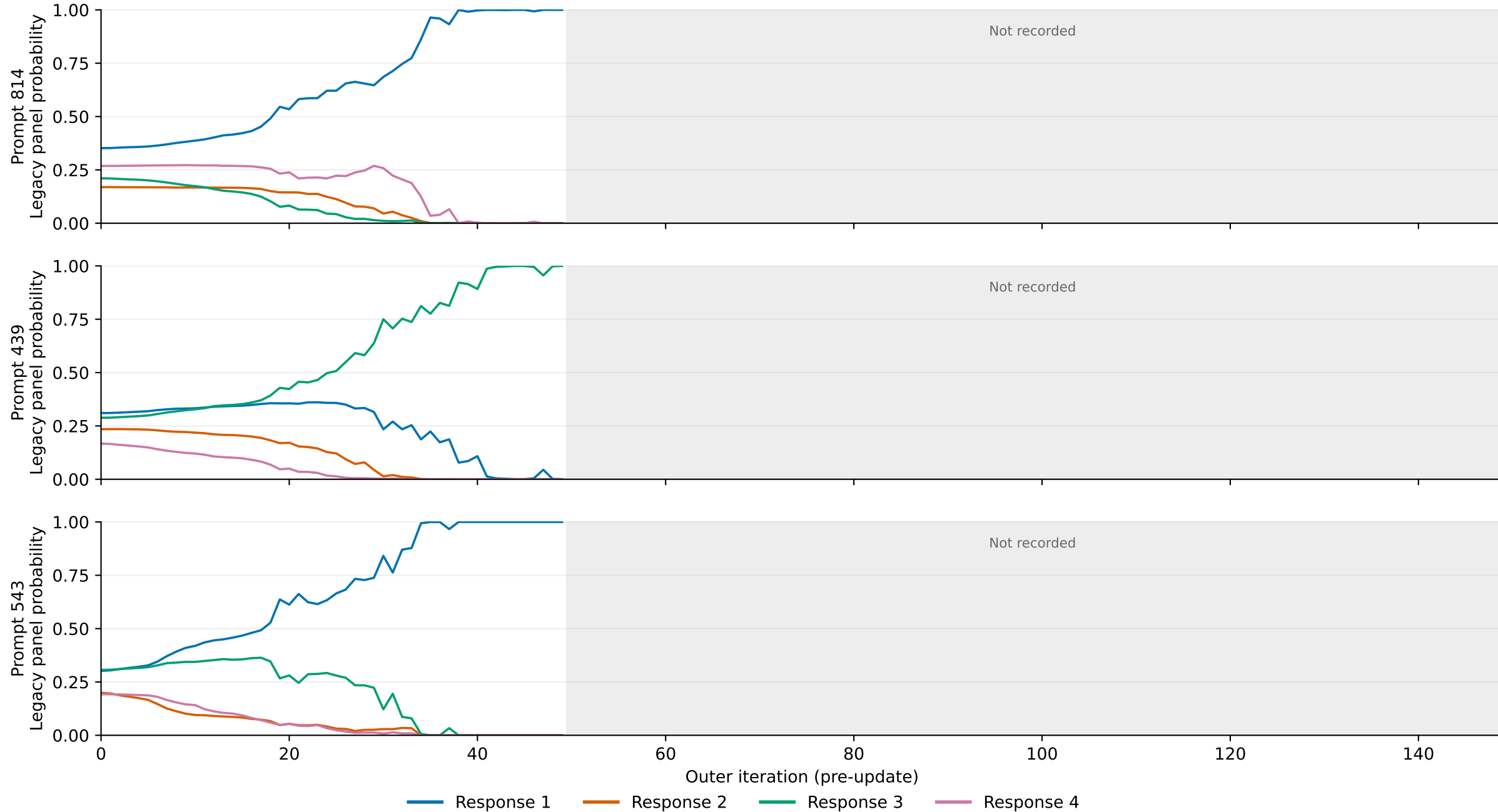
Response 3

Response 4

DPO | Original cyclic experiments

Legacy definition: $\pi_i = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π_i

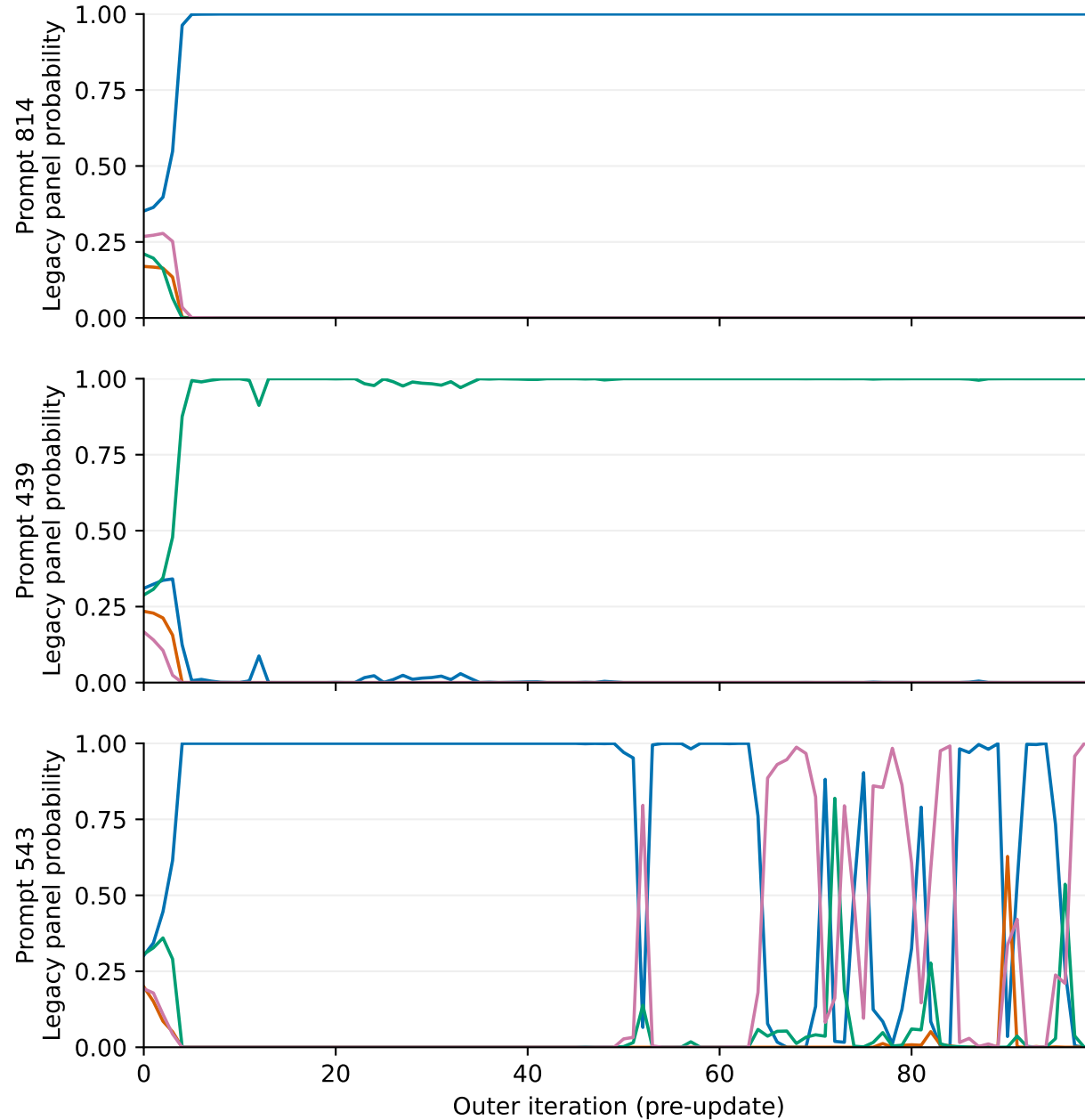
$\alpha=0$, $\lambda=0$, $\beta=0.01$



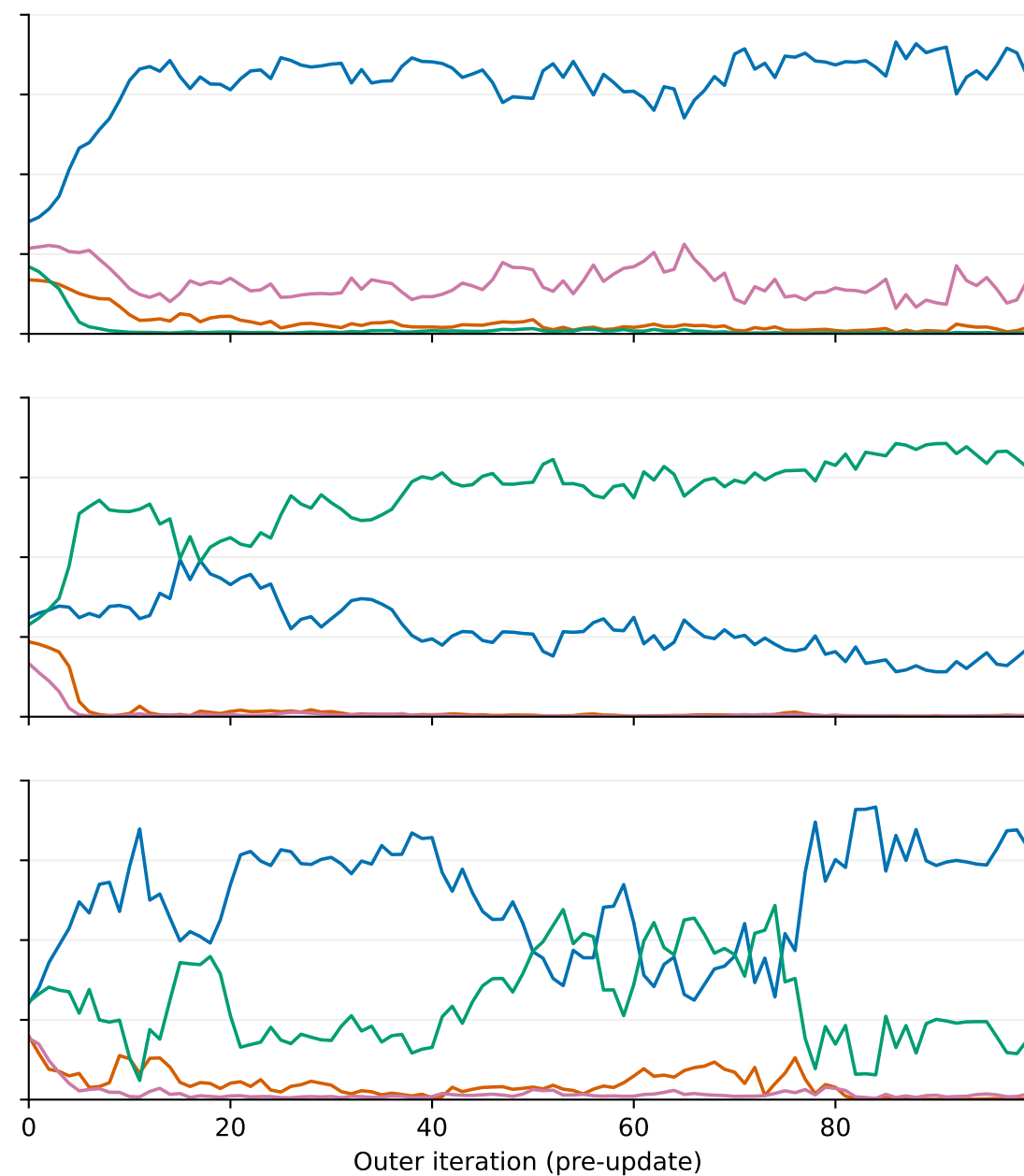
IPO | Fixed $\alpha=.99$, $\lambda=.8$: beta comparison

Legacy definition: $\pi = \text{softmax}(\text{response-token-average log probability})$; not sequence-sum π

$\alpha=0.99$, $\lambda=0.8$, $\beta=1$



$\alpha=0.99$, $\lambda=0.8$, $\beta=10$



Response 1 Response 2 Response 3 Response 4